

DAY 25 | Deep Dive into Linear Regression

60-Day Data Science Learning Journey • Muhammad Ahmad

Learning Objectives

Understand Linear Regression beyond calling `.fit()`: the equation, slope, intercept, coefficients, predictions, residuals, MSE, MAE, R^2 , best-fitting line, multiple regression, assumptions, overfitting, evaluation, and a realistic student-performance project.

1. What is Linear Regression?

A supervised Machine Learning algorithm mainly used to predict numerical values such as marks, salary, price, or sales.

2. Regression Equation

For one feature: $\hat{y} = b_0 + b_1x$. $\hat{y}$ is prediction, b_0 is intercept, b_1 is slope/coefficient, and x is the feature.

3. Slope and Intercept

The slope represents the estimated change in prediction for a one-unit increase in the feature. The intercept is the predicted value when x is zero, although that interpretation is not always meaningful.

4. Simple vs Multiple Regression

Simple regression uses one feature. Multiple regression uses several: $\hat{y} = b_0 + b_1x_1 + \dots + b_nx_n$. Coefficients in multiple regression are interpreted conditional on the other included predictors.

5. Residuals

Residual = Actual – Predicted. Positive residual means the model predicted too low. Negative residual means it predicted too high.

6. Why Square Errors?

Positive and negative errors can cancel. Squaring prevents cancellation and gives larger errors more influence. Ordinary Least Squares minimizes the sum of squared residuals.

7. Evaluation Metrics

MSE is average squared error. MAE is average absolute error in target units. R^2 compares the model with a mean-prediction baseline.

8. Scikit-learn Coefficients

After fitting, `model.coef_` contains learned coefficients and `model.intercept_` contains the intercept.

9. Regression Assumptions

Consider linearity, independence where appropriate, residual variance, residual patterns, multicollinearity, and influential outliers.

10. Overfitting

Excellent training performance with weak test performance can indicate poor generalization. The goal is useful performance on unseen data.

Complete Project | Student Performance Prediction System

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

data = {
    "study_hours": [1,2,3,4,5,6,7,8,2,4,6,8,3,5,7,9,1,10,9,10],
    "attendance": [60,65,70,72,75,78,82,85,68,74,80,88,71,76,84,62,90,95,92,96],
    "sleep_hours": [5,5.5,6,6,6.5,7,7,7.5,5.5,6,7,8,6,6.5,7,5,8,8,7.5,8],
    "previous_gpa": [2.1,2.2,2.4,2.5,2.7,2.8,3.0,3.1,2.3,2.6,2.9,3.2,2.5,2.7,3.0,2.0,3.4,3.6,3.3,3.7],
    "exam_marks": [32,40,48,55,64,70,79,86,43,58,72,89,50,63,78,34,92,96,91,98]
}

df = pd.DataFrame(data)
X = df[["study_hours", "attendance", "sleep_hours", "previous_gpa"]]
y = df["exam_marks"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("Model Coefficients:")
for feature, coefficient in zip(X.columns, model.coef_):
    print(feature, ":", round(coefficient, 4))
print("Intercept:", round(model.intercept_, 4))
print("MAE:", round(mae, 2))
print("MSE:", round(mse, 2))
print("R²:", round(r2, 3))

new_student = pd.DataFrame({"study_hours": [6.5], "attendance": [85], "sleep_hours": [7], "previous_gpa": [3.1]})
prediction = model.predict(new_student)
print("Predicted marks:", round(prediction[0], 2))

plt.scatter(y_test, y_pred)
plt.xlabel("Actual Marks")
plt.ylabel("Predicted Marks")
plt.title("Actual vs Predicted Exam Marks")
plt.tight_layout()
plt.show()

residuals = y_test - y_pred
plt.scatter(y_pred, residuals)
plt.axhline(0, linestyle="--")
plt.xlabel("Predicted Values")
plt.ylabel("Residuals")
plt.title("Residual Plot")
plt.tight_layout()
plt.show()
```

Line-by-Line Explanation

Imports: Pandas handles tabular data, Matplotlib visualizes results, and Scikit-learn provides splitting, regression, and metrics.

X and y: X contains the four features. y contains exam_marks, the numerical target.

Train/Test Split: 80% is used for training and 20% for testing. random_state makes the split reproducible.

Training: model.fit learns coefficients and an intercept from the training data.

Prediction: model.predict generates predictions for unseen test rows and later for a new student.

Metrics: MAE, MSE, and R² summarize different aspects of prediction performance.

Residual Plot: Residuals can reveal systematic patterns that a single score may hide.

Practice Tasks + Complete Solutions

Task 1: Train Linear Regression using hours [1,2,3,4,5] and marks [35,45,52,65,75].

```
X=[[1],[2],[3],[4],[5]]
y=[35,45,52,65,75]
model=LinearRegression()
model.fit(X,y)
```

Task 2: Print coefficient and intercept.

```
print(model.coef_)
print(model.intercept_)
```

Task 3: Predict marks for 6 study hours.

```
print(model.predict([[6]])[0])
```

Task 4: Calculate MAE, MSE and R².

```
mae=mean_absolute_error(y_test,y_pred)
mse=mean_squared_error(y_test,y_pred)
r2=r2_score(y_test,y_pred)
print(mae,mse,r2)
```

Task 5: Use study_hours and attendance as two features.

```
x=df[["study_hours","attendance"]]
y=df["exam_marks"]
model=LinearRegression().fit(X,y)
```

Task 6: Create a residual plot.

```
residuals=y_test-y_pred
plt.scatter(y_pred,residuals)
plt.axhline(0,linestyle="--")
plt.show()
```

Task 7: Create an Actual vs Predicted plot.

```
plt.scatter(y_test,y_pred)
plt.xlabel("Actual")
plt.ylabel("Predicted")
plt.show()
```

Task 8: Compare test_size 0.2 and 0.3.

```
for size in [0.2,0.3]:
    X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=size,random_state=42)
    model=LinearRegression().fit(X_train,y_train)
    print(size,r2_score(y_test,model.predict(X_test)))
```

Day 25 Quick Revision

Core equation: $y = b_0 + b_1x$

Multiple regression: $y = b_0 + b_1x_1 + \dots + b_nx_n$

Residual: Actual – Predicted

MAE: average absolute error

MSE: average squared error

R²: comparison with a mean-prediction baseline

Goal: useful generalization to unseen data.

Important: the student dataset is educational and small. It should not be used for real academic decisions.